

ENGR 213: Applied Ordinary Differential Equations

Past Midterm Examinations Solved Master Guide (2011–2018)

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Curriculum Scope: First-Order ODEs, Separable Equations, Integrating Factors, Exact Equations, Bernoulli & Homogeneous Substitutions, Linear Models & Kinematics, Higher-Order Linear ODEs, Linear Independence & Wronskians, Method of Undetermined Coefficients, Variation of Parameters, and Cauchy-Euler Equations.

Pedagogical Architecture & Exam Strategy

Every problem in this compendium is extracted directly from official Concordia University ENGR 213 Midterm Examinations (2011–2018). Each solution follows our rigorous standard:

1. **Explicit Step Breakdown:** Every phase of the derivation is explicitly titled (**Step 1** , **Step 2** ...).
 2. **Theory Before Algebra:** The mathematical theorem, standard form, or substitution rule is stated before executing calculations.
 3. **Zero Skipped Calculations:** All integrals, derivatives, partial derivatives, and fraction simplifications are displayed line-by-line.
 4. **Exam Traps & Verification:** Common pitfalls that result in lost marks on Concordia exams are explicitly highlighted with cross-checks.
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Part I: Fall 2018 Midterm Examination 1

Problem 1: Separable Initial Value Problem

Problem Statement

Solve the initial value problem for the first-order differential equation:

$$\frac{dy}{dx} = \frac{3x^2y}{1+x^3}, \quad y(1) = 2$$

Complete Step-by-Step Solution

Step 1: Separation of Variables

Move all terms in y to the left-hand side and all terms in x to the right-hand side:

$$\frac{1}{y} dy = \frac{3x^2}{1+x^3} dx$$

Step 2: Integration of Both Sides

Integrate both sides independently:

$$\int \frac{1}{y} dy = \int \frac{3x^2}{1+x^3} dx$$

On the right-hand side, substitute $u = 1 + x^3$, so $du = 3x^2 dx$:

$$\int \frac{du}{u} = \ln |u| + C_1 = \ln |1 + x^3| + C_1$$

On the left-hand side:

$$\ln |y| = \ln |1 + x^3| + C_1$$

Step 3: Exponentiation

Exponentiate both sides with base e :

$$|y| = e^{\ln|1+x^3|+C_1} = e^{C_1}|1+x^3|$$

Letting $C = \pm e^{C_1}$:

$$y(x) = C(1+x^3)$$

Step 4: Apply Initial Condition $y(1) = 2$

Substitute $x = 1$ and $y = 2$:

$$2 = C(1+1^3) = 2C \implies C = 1$$

Step 5: Final Explicit Solution

$$y(x) = 1+x^3 \quad \text{for } x > -1$$

Exam Trap Alert: Writing $\ln|y| = \ln|1+x^3| + C \implies y = 1+x^3 + C$ is an instant zero on Concordia exams! The constant C enters the exponent, meaning $y = C(1+x^3)$.

Problem 2: Bernoulli Differential Equation

Problem Statement

Solve the first-order differential equation:

$$\frac{dy}{dx} + 2xy = -xy^4$$

Complete Step-by-Step Solution

Step 1: Recognize Bernoulli Form

The equation matches standard Bernoulli form:

$$y' + P(x)y = Q(x)y^n \quad \text{with } P(x) = 2x, Q(x) = -x, n = 4$$

Step 2: Divide by $y^n = y^4$

$$y^{-4} \frac{dy}{dx} + 2xy^{-3} = -x$$

Step 3: Linearizing Substitution

Let $u = y^{1-n} = y^{1-4} = y^{-3}$.

Differentiating with respect to x via the chain rule:

$$\frac{du}{dx} = -3y^{-4} \frac{dy}{dx} \implies y^{-4} \frac{dy}{dx} = -\frac{1}{3} \frac{du}{dx}$$

Substitute into the differential equation:

$$-\frac{1}{3} \frac{du}{dx} + 2xu = -x$$

Step 4: Put into Standard Linear Form

Multiply by -3 :

$$\frac{du}{dx} - 6xu = 3x$$

Step 5: Integrating Factor Method

Compute integrating factor $\mu(x)$:

$$\mu(x) = e^{\int -6x, dx} = e^{-3x^2}$$

Multiply the linear ODE by $\mu(x)$:

$$\begin{aligned} e^{-3x^2} \frac{du}{dx} - 6xe^{-3x^2} u &= 3xe^{-3x^2} \\ \frac{d}{dx} [e^{-3x^2} u] &= 3xe^{-3x^2} \end{aligned}$$

Step 6: Integration

Integrate both sides with respect to x :

$$e^{-3x^2} u = \int 3xe^{-3x^2}, dx$$

Let $w = -3x^2 \implies dw = -6x, dx \implies 3x, dx = -\frac{1}{2}, dw$:

$$\int 3xe^{-3x^2}, dx = -\frac{1}{2} \int e^w, dw = -\frac{1}{2} e^{-3x^2} + C$$

Therefore:

$$e^{-3x^2} u = -\frac{1}{2} e^{-3x^2} + C \implies u(x) = -\frac{1}{2} + Ce^{3x^2}$$

Step 7: Back-Substitute $u = y^{-3}$

$$y^{-3} = -\frac{1}{2} + Ce^{3x^2} \iff \frac{1}{y^3} = \frac{2Ce^{3x^2} - 1}{2}$$

$$y(x) = \left(-\frac{1}{2} + Ce^{3x^2} \right)^{-1/3}$$

Problem 3: Exact First-Order Differential Equation

Problem Statement

Solve the differential equation:

$$(4x^3 + 3x^2 + 3y), dx + (3x + 2y + 1), dy = 0$$

Complete Step-by-Step Solution

Step 1: Identify $M(x, y)$ **and** $N(x, y)$

$$M(x, y) = 4x^3 + 3x^2 + 3y$$

$$N(x, y) = 3x + 2y + 1$$

Step 2: Test for Exactness

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(4x^3 + 3x^2 + 3y) = 3$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(3x + 2y + 1) = 3$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 3$, the equation is **exact**.

Step 3: Integrate $M(x, y)$ **with Respect to** x

A potential function $F(x, y)$ exists such that $\frac{\partial F}{\partial x} = M$:

$$F(x, y) = \int (4x^3 + 3x^2 + 3y), dx = x^4 + x^3 + 3xy + g(y)$$

Step 4: Differentiate $F(x, y)$ with Respect to y and Equate to N

$$\frac{\partial F}{\partial y} = 0 + 0 + 3x + g'(y) = 3x + g'(y)$$

Equating to $N(x, y) = 3x + 2y + 1$:

$$3x + g'(y) = 3x + 2y + 1 \implies g'(y) = 2y + 1$$

Step 5: Integrate $g'(y)$

$$g(y) = \int (2y + 1) dy = y^2 + y$$

Step 6: Construct Potential Function and Final Implicit Solution

$$F(x, y) = x^4 + x^3 + 3xy + y^2 + y$$

The general solution is:

$$x^4 + x^3 + 3xy + y^2 + y = C$$

Problem 4: Homogeneous Degree-3 Equation IVP

Problem Statement

Solve the initial value problem:

$$(x^3 + y^3), dx - 3xy^2, dy = 0, \quad y(1) = 1$$

Complete Step-by-Step Solution

Step 1: Verify Homogeneity

$$M(tx, ty) = t^3x^3 + t^3y^3 = t^3M(x, y)$$

$$N(tx, ty) = -3(tx)(t^2y^2) = t^3N(x, y)$$

Both coefficients are homogeneous of degree 3.

Step 2: Apply Substitution $y = vx$

$$y = vx \implies dy = v, dx + x, dv$$

Substitute into the ODE:

$$(x^3 + v^3x^3), dx - 3x(v^2x^2)(v, dx + x, dv) = 0$$

Step 3: Divide by x^3 ($x \neq 0$)

$$(1 + v^3), dx - 3v^2(v, dx + x, dv) = 0$$

$$(1 + v^3 - 3v^3), dx - 3v^2x, dv = 0$$

$$(1 - 2v^3), dx = 3v^2x, dv$$

Step 4: Separate Variables

$$\frac{1}{x}, dx = \frac{3v^2}{1 - 2v^3}, dv$$

Step 5: Integrate Both Sides

$$\int \frac{1}{x}, dx = \int \frac{3v^2}{1 - 2v^3}, dv$$

On the right-hand side, substitute $w = 1 - 2v^3 \implies dw = -6v^2, dv \implies 3v^2, dv = -\frac{1}{2}, dw$:

$$\ln |x| = -\frac{1}{2} \int \frac{dw}{w} = -\frac{1}{2} \ln |1 - 2v^3| + \ln C_1$$

$$\ln |x| + \frac{1}{2} \ln |1 - 2v^3| = \ln C_1$$

Multiply by 2:

$$2 \ln |x| + \ln |1 - 2v^3| = 2 \ln C_1 = \ln C$$

$$\ln |x^2(1 - 2v^3)| = \ln C \implies x^2(1 - 2v^3) = C$$

Step 6: Back-Substitute $v = \frac{y}{x}$

$$x^2 \left(1 - 2 \frac{y^3}{x^3} \right) = C \implies x^2 - \frac{2y^3}{x} = C \implies x^3 - 2y^3 = Cx$$

Step 7: Apply Initial Condition $y(1) = 1$

$$1^3 - 2(1^3) = C(1) \implies 1 - 2 = C \implies C = -1$$

Substitute $C = -1$:

$$x^3 - 2y^3 = -x \iff 2y^3 = x^3 + x$$

$$\boxed{y(x) = \sqrt[3]{\frac{x^3 + x}{2}}}$$

Problem 5: Bacterial Growth Kinetics & Tripling Period

Problem Statement

In a culture of bacteria, the rate of increase is proportional to the number present. If it is found that the number triples in 4 hours, how many times the initial population may be expected at the end of 12 hours?

Complete Step-by-Step Solution

Step 1: Mathematical Model Formulation

Let $x(t)$ denote the population at time t (in hours), and $x_0 = x(0)$ be the initial population:

$$\frac{dx}{dt} = kx$$

Separating and integrating:

$$\int \frac{dx}{x} = \int k, dt \implies \ln |x| = kt + \ln C \implies x(t) = x_0 e^{kt}$$

Step 2: Apply the Tripling Condition at $t = 4$ hours

$$x(4) = 3x_0$$

$$x_0 e^{4k} = 3x_0 \implies e^{4k} = 3$$

Step 3: Evaluate at $t = 12$ hours

Notice that $12 = 3 \times 4$. Using properties of exponents:

$$x(12) = x_0 e^{12k} = x_0 (e^{4k})^3$$

Substitute $e^{4k} = 3$:

$$x(12) = x_0 (3)^3 = 27x_0$$

Step 4: Final Conclusion

At the end of 12 hours, the population will be 27 times the original amount.

Part II: Winter 2016 Midterm Examination II (Version A)

Problem 1: Linear Independence via Wronskian & Trig Identities

Problem Statement

Determine which system of functions is linearly independent and which is linearly dependent on $(0, \infty)$:

1. $S_1 = \left\{ 2, \frac{1}{x^2}, \frac{\ln x}{x^2} \right\}$
2. $S_2 = \left\{ \sin x, \cos(2x), 1 - \sin x - 2\sin^2 x \right\}$

Complete Step-by-Step Solution

Analysis of S_1 :

Let $f_1(x) = 2$, $f_2(x) = x^{-2}$, $f_3(x) = x^{-2} \ln x$. Compute the Wronskian:

$$W(x) = \begin{vmatrix} 2 & x^{-2} & x^{-2} \ln x \\ 0 & -2x^{-3} & x^{-3}(1 - 2 \ln x) \\ 0 & 6x^{-4} & -x^{-4}(5 - 6 \ln x) \end{vmatrix}$$

Expand along the first column:

$$W(x) = 2 \begin{vmatrix} -2x^{-3} & x^{-3}(1 - 2 \ln x) \\ 6x^{-4} & -x^{-4}(5 - 6 \ln x) \end{vmatrix}$$

$$W(x) = 2 [(-2x^{-3})(-x^{-4}(5 - 6 \ln x)) - (6x^{-4})(x^{-3}(1 - 2 \ln x))]$$

$$W(x) = 2 [2x^{-7}(5 - 6 \ln x) - 6x^{-7}(1 - 2 \ln x)]$$

$$W(x) = 2x^{-7}[10 - 12 \ln x - 6 + 12 \ln x] = 2x^{-7}(4) = 8x^{-7} > 0 \quad \forall x > 0$$

Since $W(x) \neq 0$, S_1 is linearly independent.

Analysis of S_2 :

Let $f_1 = \sin x$, $f_2 = \cos(2x)$, $f_3 = 1 - \sin x - 2\sin^2 x$.

Recall the double-angle trigonometric identity:

$$\cos(2x) = 1 - 2\sin^2 x$$

Substitute into f_3 :

$$f_3(x) = (1 - 2\sin^2 x) - \sin x = \cos(2x) - \sin x = f_2(x) - f_1(x)$$

Rearranging:

$$f_1(x) - f_2(x) + f_3(x) = 0 \quad \text{for all } x$$

Because a non-trivial linear combination vanishes identically ($c_1 = 1$, $c_2 = -1$, $c_3 = 1$), S_2 is linearly dependent.

Problem 2: Fourth-Order Linear ODE with Repeated Real Roots

Problem Statement

Find the general solution of the differential equation:

$$16y^{(4)} - 72y'' + 81y = 0$$

Complete Step-by-Step Solution

Step 1: Formulate the Auxiliary Equation

Let $y = e^{mx}$. Substituting gives:

$$16m^4 - 72m^2 + 81 = 0$$

Step 2: Factor as a Quadratic in m^2

$$(4m^2 - 9)^2 = 0$$

Factor each difference of squares:

$$[(2m - 3)(2m + 3)]^2 = (2m - 3)^2(2m + 3)^2 = 0$$

Step 3: Identify Roots and Multiplicities

- $m_1 = m_2 = \frac{3}{2}$ (real root, multiplicity 2)
- $m_3 = m_4 = -\frac{3}{2}$ (real root, multiplicity 2)

Step 4: Construct Fundamental Set of Solutions

- For $m = \frac{3}{2}$: $y_1(x) = e^{\frac{3}{2}x}$, $y_2(x) = xe^{\frac{3}{2}x}$
- For $m = -\frac{3}{2}$: $y_3(x) = e^{-\frac{3}{2}x}$, $y_4(x) = xe^{-\frac{3}{2}x}$

Step 5: General Solution

$$y(x) = (C_1 + C_2x)e^{\frac{3}{2}x} + (C_3 + C_4x)e^{-\frac{3}{2}x}$$

Problem 3: Resonant Undetermined Coefficients IVP

Problem Statement

Use the method of undetermined coefficients to solve the initial value problem:

$$y'' - 4y' = 2xe^{4x}, \quad y(0) = -\frac{1}{4}, \quad y'(0) = \frac{1}{8}$$

Complete Step-by-Step Solution

Step 1: Solve the Homogeneous Equation

$$m^2 - 4m = m(m - 4) = 0 \implies m_1 = 0, m_2 = 4$$

$$y_c(x) = C_1 + C_2 e^{4x}$$

Step 2: Set Up Particular Solution Candidate $y_p(x)$

The forcing function is $g(x) = (2x)e^{4x}$. Normally, candidate would be $(Ax + B)e^{4x}$. However, e^{4x} is already present in $y_c(x)$ (resonance). Multiply candidate by x :

$$y_p(x) = x(Ax + B)e^{4x} = (Ax^2 + Bx)e^{4x}$$

Step 3: Compute Derivatives

$$y'_p(x) = (2Ax + B)e^{4x} + 4(Ax^2 + Bx)e^{4x} = e^{4x} [4Ax^2 + (2A + 4B)x + B]$$

$$y''_p(x) = e^{4x} [16Ax^2 + (16A + 16B)x + 2A + 8B]$$

Step 4: Substitute into $y'' - 4y'$

$$y''_p - 4y'_p = e^{4x} [8Ax + 2A + 4B] \equiv 2xe^{4x}$$

Equate coefficients:

$$8A = 2 \implies A = \frac{1}{4}$$

$$2A + 4B = 0 \implies 2\left(\frac{1}{4}\right) + 4B = 0 \implies 4B = -\frac{1}{2} \implies B = -\frac{1}{8}$$

Therefore:

$$y_p(x) = e^{4x} \left(\frac{1}{4}x^2 - \frac{1}{8}x \right)$$

Step 5: General Solution

$$y(x) = C_1 + C_2 e^{4x} + e^{4x} \left(\frac{1}{4}x^2 - \frac{1}{8}x \right)$$

Step 6: Apply Initial Conditions

$$y(0) = C_1 + C_2 = -\frac{1}{4}$$

Differentiate $y(x)$:

$$y'(x) = 4C_2 e^{4x} + 4e^{4x} \left(\frac{1}{4}x^2 - \frac{1}{8}x \right) + e^{4x} \left(\frac{1}{2}x - \frac{1}{8} \right)$$

At $x = 0$:

$$y'(0) = 4C_2 - \frac{1}{8} = \frac{1}{8} \implies 4C_2 = \frac{2}{8} = \frac{1}{4} \implies C_2 = \frac{1}{16}$$
$$C_1 = -\frac{1}{4} - C_2 = -\frac{4}{16} - \frac{1}{16} = -\frac{5}{16}$$

Step 7: Final Unique Solution

$$y(x) = e^{4x} \left(\frac{1}{4}x^2 - \frac{1}{8}x + \frac{1}{16} \right) - \frac{5}{16}$$

Problem 4: Variation of Parameters Boundary Value Problem

Problem Statement

Using the method of variation of parameters, solve the boundary value problem:

$$\frac{d^2y}{dx^2} + y = 2 \sec^3 x, \quad y(0) = -2, \quad y\left(\frac{\pi}{4}\right) = 0$$

Complete Step-by-Step Solution

Step 1: Homogeneous Solution and Wronskian

$$m^2 + 1 = 0 \implies m = \pm i \implies y_1 = \cos x, y_2 = \sin x$$

$$W(y_1, y_2) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

Step 2: Formulas for $u_1'(x)$ and $u_2'(x)$

$$u_1'(x) = -\frac{y_2 f(x)}{W} = -\frac{\sin x (2 \sec^3 x)}{1} = -2 \frac{\sin x}{\cos^3 x}$$

$$u_2'(x) = \frac{y_1 f(x)}{W} = \frac{\cos x (2 \sec^3 x)}{1} = 2 \sec^2 x$$

Step 3: Integrate to Find $u_1(x)$ and $u_2(x)$

$$u_1(x) = -2 \int \frac{\sin x}{\cos^3 x} dx = 2 \int (\cos x)^{-3} d(\cos x) = -\frac{1}{\cos^2 x} = -\sec^2 x$$

$$u_2(x) = 2 \int \sec^2 x dx = 2 \tan x$$

Step 4: Particular Solution $y_p(x)$

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x) = (-\sec^2 x)(\cos x) + (2 \tan x)(\sin x)$$

$$y_p(x) = -\sec x + 2 \sin x \tan x$$

Step 5: General Solution

$$y(x) = C_1 \cos x + C_2 \sin x + 2 \sin x \tan x - \sec x$$

Step 6: Apply Boundary Conditions

At $x = 0$:

$$y(0) = C_1(1) + C_2(0) + 0 - 1 = -2 \implies C_1 - 1 = -2 \implies C_1 = -1$$

At $x = \frac{\pi}{4}$:

$$y\left(\frac{\pi}{4}\right) = (-1)\frac{\sqrt{2}}{2} + C_2\frac{\sqrt{2}}{2} + 2\left(\frac{\sqrt{2}}{2}\right)(1) - \sqrt{2} = 0$$

$$\frac{\sqrt{2}}{2}(C_2 - 1) + \sqrt{2} - \sqrt{2} = 0 \implies C_2 = 1$$

Step 7: Final Solution

$$y(x) = \sin x - \cos x + 2 \sin x \tan x - \sec x$$

Problem 5: Cauchy-Euler Complex Conjugate Equation**Problem Statement**

Solve the Cauchy-Euler differential equation:

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 13y = 0, \quad x > 0$$

Complete Step-by-Step Solution

Step 1: Substitute $y = x^m$

$$y' = mx^{m-1}, \quad y'' = m(m-1)x^{m-2}$$

$$x^2[m(m-1)x^{m-2}] - 3x[mx^{m-1}] + 13x^m = 0$$

$$x^m [m(m-1) - 3m + 13] = 0$$

Step 2: Auxiliary Equation

$$m^2 - 4m + 13 = 0$$

Complete the square:

$$(m - 2)^2 + 9 = 0 \implies (m - 2)^2 = -9 \implies m = 2 \pm 3i$$

Step 3: Complex Roots Rule for Cauchy-Euler

For roots $m = \alpha \pm i\beta$, the basis solutions are $x^\alpha \cos(\beta \ln x)$ and $x^\alpha \sin(\beta \ln x)$.

Here $\alpha = 2$ and $\beta = 3$:

$$y(x) = x^2 [C_1 \cos(3 \ln x) + C_2 \sin(3 \ln x)]$$

Bonus Problem: Asymptotic Stability at Infinity

Problem Statement

For which real values of parameters a and b do all solutions of $y'' + ay' + by = 0$ approach 0 as $x \rightarrow \infty$?

Complete Step-by-Step Solution

The auxiliary equation is $m^2 + am + b = 0$, with roots:

$$m_{1,2} = -\frac{a}{2} \pm \sqrt{\frac{a^2}{4} - b}$$

1. **Case 1: Complex Roots** ($b > \frac{a^2}{4}$):

$$y(x) = e^{-\frac{a}{2}x} [C_1 \cos(\omega x) + C_2 \sin(\omega x)]$$

Approaches 0 as $x \rightarrow \infty$ if and only if the exponential decay rate is strictly positive: $-\frac{a}{2} < 0 \iff a > 0$.
Since $b > a^2/4 > 0$, $b > 0$.

2. **Case 2: Repeated Real Root** ($b = \frac{a^2}{4}$):

$$y(x) = (C_1 + C_2 x) e^{-\frac{a}{2}x}$$

By L'Hôpital's Rule, $\lim_{x \rightarrow \infty} x e^{-\frac{a}{2}x} = 0$ if and only if $a > 0$. Since $b = a^2/4$, $b > 0$.

3. **Case 3: Distinct Real Roots** ($0 < b < \frac{a^2}{4}$): Both roots must be strictly negative:

$$m_1 = -\frac{a}{2} + \sqrt{\frac{a^2}{4} - b} < 0 \iff \sqrt{\frac{a^2}{4} - b} < \frac{a}{2}$$

Squaring requires $a > 0$ and $\frac{a^2}{4} - b < \frac{a^2}{4} \iff b > 0$.

Conclusion

$$a > 0 \quad \text{and} \quad b > 0$$

Part III: Winter 2016 Midterm Examination II (Version B)

Problem 1: Trigonometric Identity Dependence Analysis

Problem Statement

Determine if $S = \{1, \sin(2x), (\sin x - \cos x)^2\}$ is linearly independent or dependent.

Solution

Expand the third function:

$$f_3(x) = (\sin x - \cos x)^2 = \sin^2 x - 2 \sin x \cos x + \cos^2 x$$

Recall $\sin^2 x + \cos^2 x = 1$ and $2 \sin x \cos x = \sin(2x)$:

$$f_3(x) = 1 - \sin(2x) = f_1(x) - f_2(x)$$

Rearranging gives $f_1(x) - f_2(x) - f_3(x) = 0$, so S is linearly dependent.

Problem 2: Fourth-Order Linear ODE (Real & Imaginary Roots)

Problem Statement

Find the general solution of $y^{(4)} - 16y = 0$.

Solution

Auxiliary equation:

$$m^4 - 16 = 0 \implies (m^2 - 4)(m^2 + 4) = 0 \implies (m - 2)(m + 2)(m - 2i)(m + 2i) = 0$$

Roots: $m = 2, -2, 2i, -2i$.

$$y(x) = C_1 e^{2x} + C_2 e^{-2x} + C_3 \cos(2x) + C_4 \sin(2x)$$

Part IV: Sample Midterm Examination II

Problem 1: Polynomial Basis Linear Dependence

Problem Statement

Determine whether $f_1 = 1, f_2 = x, f_3 = x^2, f_4 = x^3, f_5 = 5x^3 + (x - 2)^2$ are linearly independent on $(0, \infty)$.

Solution

Expand $f_5(x)$:

$$f_5(x) = 5x^3 + x^2 - 4x + 4 = 5f_4(x) + f_3(x) - 4f_2(x) + 4f_1(x)$$

Since f_5 is an explicit linear combination of the first four polynomials, The set is linearly dependent.